

**INDIAN INSTITUTE OF TECHNOLOGY
KHARAGPUR**

1. Name of the Academic Unit: Centre of Excellence in Artificial Intelligence

2. Subject Name: Deep Learning Laboratory for Sequential Data

L-T-P: 0-0-3

Credits: 2

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

Predictive Models for text and image based on RNN, word embeddings for NLP, applications of sentence completion, translation. Autoregressive based Seq-to-seq prediction. Modeling long-range sequential relations using LSTM, GRU and Attention networks. ConvLSTMs and TrajGRUs for spatio-temporal prediction. Simple LLMs based on Transformers. Speech and multi-modal sequential data. Study of ChatGPT and other popular generative models for text

Reference Books:

1) Generative AI with Python and TensorFlow 2: Create images, text, and music with VAEs, GANs, LSTMs, Transformer models by Babcock and Bali, Packt Publishing (2021)

1) Introduction to Generative AI by Dhamani and Engler (Not published yet)

2) Technical reports and documentations of ChatGPT, GPT-4, Dall-E, etc

3) Relevant research papers

4) <https://github.com/ashishpatel26/Andrew-NG-Notes/blob/master/andrewng-p-5-sequence-models.md>

5. Lecture-wise break-up:

Sl. No.	Topic	No. of Lab-Hours
1. RNN for Time-series Forecasting	Simple Recurrent Neural Network for time-series prediction (using simulated and stock-market data)	3 hours
2. Word Embeddings	Creating text sequences using word embeddings (GLoVe and Word2Vec), sentence autocompletion using RNN	3 hours

3. Neural Image Generation	Image generation using PixelCNN, PixelRNN	3 hours
4. Sequence-to-Sequence	Encoder-decoder approach, autoregressive sequential prediction vs beam search	3 hours
5. Limitations of RNN	Seq2Seq approach for neural language translation, evaluating long-term sequential dependency modeling in text data, understanding vanishing and exploding gradients	3 hours
6. LSTM and GRU	Seq2Seq approach for neural language translation (including long-term dependency) using LSTM and GRU	3 hours
7. Spatio-temporal Prediction	ConvLSTM and TrajGRU for spatio-temporal sequence prediction (rainfall map prediction)	3 hours
8. Attention Model	Neural language translation: Understanding word-word relations using Attention Networks, self-attention and multi-head attention	3 hours
9. Large Language Models	Text Generation using Transformers	3 hours
10. Multimodal data generation	image from text prompt, textual summarization of image	3hours
11. Speech data modeling	Modeling speech/audio sequences using above models, audio feature extraction, speech recognition and trigger word detection	3 hours
12. ChatGPT and Prompt Engineering	Study of ChatGPT, GPT-4, other public generative models for text, prompt Engineering for complex outputs from these models	3 hours
Total number of hours		36